

Naziru HALILU

DIGITAL AGRICULTURE

CONTACT

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ORCID: 0009-0006-2324-9599

Zenodo: publications and data

Website: personal portfolio

DETAILS

Location: Pamplona, Spain

Nationality: Nigerian

EDUCATION

2025 to expected Jun/Aug 2027

EMJMD, Cutting-Edge Technologies for Sustainable Agriculture (SUSTAGRI)

UPNA, UTAD, AUA

2019 to 2024

B.Eng, Agric. and Bioresources Engineering

ABU, Nigeria. First Class Honours, CGPA 4.77 of 5.00, Top 1 percent

2016 to 2018

National Diploma, Agric. Engineering Technology

ABU. Distinction, CGPA 3.78 of 4.00, Top 1 percent

SKILLS

Python / Deep Learning

GIS and Remote Sensing

Computer Vision

Geostatistics

Agric. Machinery Systems

MATLAB / R

LANGUAGES

English (fluent)

MEMBERSHIPS

ASABE, NIAE, NSE, Pupil Engineer COREN

PROFILE

Agricultural engineer with expertise across digital agriculture (computer vision, geospatial machine learning, and precision farming), specialized in building phenotyping, crop-stress, erosion, and predictive biophysical modeling systems that turn raw field signals into decisions before problems become visible.

EMPLOYMENT HISTORY

EMJMD Research Scholar

Sep 2026 to Jan 2027

Agricultural University of Athens, Greece

- Term 3 focus: efficient water use, smart irrigation (30 ECTS).
- Continued graduate level research training in smart design and management of irrigation systems.
- Ran 10,190 Daisy simulations (17 factorial experiments, ten soils) on precision irrigation and mulching, deriving a soil independent irrigation trigger law (pooled R squared 0.48 to 0.75 for irrigation, 0.23 to 0.70 for yield), a soil gated drainage controllability map (Spearman rho up to +0.95) and a rainfall conditioned mulch water balance regime map (climate and mulch interactions up to 15 percent of irrigation). DSS | Data | DOI
- Built NaCROP, a modular Python/Tkinter pipeline for evapotranspiration, soil water balance, irrigation scheduling, and a four step crop growth simulation for five smallholder crops around Zaria, Nigeria. Data

Summer Intern, Computer Vision Group

Jun to Aug 2026

Dept. of Computer Engineering, ABU, Zaria

- Accomplished cross condition transfer of crop disease recognition models, as measured by 64.4 percent classification accuracy (macro AUC 0.882 against a 5.3 percent chance baseline) across 19 disease classes on 1,543 field images, by extending the Geometry Agnostic Contrastive Learning (GACL) framework with four new geometry invariant neural components. First author paper under review at Biosystems Engineering, Elsevier. Data

EMJMD Research Scholar

Feb to Jun 2026

UTAD, Portugal

- Term 2 focus: precision agriculture for sustainable crop production (30 ECTS).
- Accomplished cross system precision agriculture prediction and input cost reduction, as measured by R squared between 0.978 and 0.995 for grape ripening and maize yield regressions and a 2.69 tonne cut in lime demand. Data
- Accomplished more reliable soil nutrient and yield mapping via Multitask Singularity Regression Kriging (MSRK), as measured by a roughly 20 percent reduction in residual kriging uncertainty across 17,864 samples. Data
- Accomplished predictive harvest date forecasting for grape ripening, as measured by R squared of 0.9784, at a six hectare UTAD vineyard. Data
- Accomplished a costed, spatially targeted pest management programme for grape moth across a 42.97 hectare Douro vineyard estate. Data

EMJMD Research Scholar

Sep 2025 to Jan 2026

Public University of Navarre, Spain

- Term 1 focus: technological bases for agricultural sustainability (30 ECTS).
- Accomplished real time, field deployable crop health screening, as measured by 33.8 percent balanced accuracy against a 4.55 percent chance baseline, by building a fourteen module RGB/multispectral vegetation indexing pipeline. Data
- Accomplished radiometrically consistent multi temporal vegetation monitoring, as measured by a 39 and 31 percent reduction in temporal NDVI variance, by designing STADOS, a five stage adaptive dark object subtraction framework. Data
- Accomplished accurate land suitability mapping for livestock slurry application, as measured by RMSE of 7.78 and R squared of 0.77 across 14.35 square kilometres near Tudela, Spain. Data
- Accomplished a 47.8 percent reduction in annual soil loss via a Bayesian grounded digital twin for gully erosion biocontrol, hold out R squared of 0.90. Data
- Built a six notebook vector/raster GIS and Sentinel 1/2 remote sensing course package in Python and Google Earth Engine. Data

Undergraduate Project Assistantships

Mar 2024 to Mar 2025

ABU, Nigeria

- Contributed data collection, statistical analysis, and literature review across concurrent research projects in water and irrigation systems, soil and erosion assessment, and postharvest and animal systems.

Project Assistant, Remote Data Lead

2024 to 2025

IAR-ABU, Nigeria

- Engineered adaptive irrigation scheduling algorithms fusing real time soil moisture and microclimate telemetry for automated greenhouse control.

Naziru Halilu

Digital Agriculture Researcher

RESEARCH IMPACT

14 Open repositories

14 Zenodo DOIs archived

4 Peer reviewed papers

6 Under review

10 Awards and honors

ORGANIZATIONS

- Agricultural University of Athens
- UTAD, Portugal
- Public University of Navarre
- Ahmadu Bello University (ABU)
- IAR-ABU
- NAERLS
- Yankari Nasarawa Farms
- BOMO Basin Project
- NCAM, Ilorin
- Takai Academy

CERTIFICATES

EMJMD certificate / enrolment

B.Eng transcript / certificate

ND transcript / certificate

REFEREES

Prof. Miguel Ángel Campo Bescós
Public University of Navarre, Spain
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Prof. José Tadeu Marques Aranha
UTAD, Portugal
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Dr. Ibrahim Babangida Dalha
Ahmadu Bello University, Nigeria
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AWARDS AND HONORS

- Erasmus Mundus Joint Master Scholarship, 2025
- Best Graduating Student, B.Eng, 2024
- Nigerian Federal Government Award for Academic Excellence, 2024
- Best Undergraduate Research Paper Award, ABU, 2024
- Best Overall Graduating Student, National Diploma, 2019
- PTDF Undergraduate Fellow, 2023

Student Research Assistant, Precision Crop Systems — 2022 to 2024

NAERLS, Nigeria

• Trained and validated CNN crop stress detection models reaching 78 percent accuracy on multispectral UAV imagery.

Project Assistant, Smart Irrigation and Water Systems — 2021 to 2022

BOMO Basin Project, Nigeria

• Implemented closed loop control logic for a one hectare smart drip irrigation array.

Research Intern — 2017 to 2020

NCAM, Ilorin and IAR, Zaria, Nigeria

• Built Python data logging pipelines to benchmark mechanization efficiency; supported baseline hydrological research.

TEACHING, LEADERSHIP AND COMMUNITY ENGAGEMENT

Undergraduate Course Tutor — 2020 to 2024

ABU Zaria (NAMET)

• Supplementary tutorials in Differential Equations and Transforms, Irrigation System Design, and Computer Applications in Agric. Engineering.

Secretary General, NAAESB — 2023 to 2024

National Assoc. of Agric. Engineering Students

• Directed operational targets and inter chapter scientific meetups for the national student body.

Director, Environmental Protection Agency — 2023 to 2024

ABU Students' Union Government

• Directed campus environmental protection and sanitation initiatives.

Deputy Coordinator — 2020 to 2025

KOJYAF, Zango Shanu, Zaria

• Coordinated tutorial and mentorship programs preparing secondary students for JAMB entrance examinations.

Physics and Mathematics Teacher — 2022 to 2023

Takai Academy, Zaria, Nigeria

• Instructed senior classes in classical physics and computational principles.

PUBLICATIONS

• **Halilu, N.**, and Sulaiman, J. (2026). Real Time RGB Proxy Vegetation Indexing and Texture Analysis for UAV and Handheld Crop Imagery. IJRSI, 13(7), 2951 to 2989. DOI

• **Halilu, N.** (2026). A Hybrid CNN-BiLSTM-Attention Deep Learning Framework for Sediment Transport in an Agricultural Gully System. IJRSI, 13(7), 2920 to 2950. DOI

• **Halilu, N.**, and Sulaiman, J. (2026). Sensing the Invisible: A Critical Review of NIR and Hyperspectral Analysis for Crop and Food Quality Assessment. IJRSI, 13(7), 2990 to 3020. DOI

• Ismail, K., and **Halilu, N.** (2024). AI Driven Precision Agriculture System for Optimizing Maize Yield in Northern Nigeria. Proc. AfCPA, 2(1), 100 to 120.

• Auwal, M., and Yusuf, A., **Naziru, H.** (2020). Physio chemical and bacteriological quality of water for irrigation purpose. Nigerian Research Journal of Engineering and Environmental Sciences, 5(1), 198 to 204.

• **Halilu, N.**, and Sulaiman, J. Geometry Agnostic Contrastive Learning (GACL). Biosystems Engineering, Elsevier. Under review. DOI

• **Halilu, N.** (2026). Digital Twin for the Evaluation of Experimental Gully Biocontrol Using Morning Glory (Ipomoea spp.). Environmental Monitoring and Assessment, Springer Nature. Under review. DOI

• **Halilu, N.**, and Sulaiman, J. Operationalizing GIS and Machine Learning across Contrasting Cropping Systems. European Journal of Agronomy, Elsevier. Under review. DOI

• **Halilu, N.** (2026). Multitask Singularity Regression Kriging for Spatio Temporal Soil NPK Crop Stress Yield Prediction. Zenodo Preprint. DOI

• **Halilu, N.** (2026). STADOS: Spectral Temporal Adaptive Dark Object Subtraction. Zenodo. DOI

• **Halilu, N.** (2025). A Systematic Review of IoT, Edge Computing, Time Series Modelling and Control Systems in Precision Irrigation. Zenodo Preprint. DOI

Manuscripts under review and in preparation

- The Reach of Suction-Triggered Irrigation and the Marginal Fate of Irrigation Water with Closed Forms from the van Genuchten-Mualem Parameters
- Soil Hydraulic Memory and the Recoverability of Evaporation Savings and the Closure of the Water Balance in a Simulation Study with Daisy
- Modeling Soil-Water-Atmosphere Interactions for Irrigation Scheduling and Design
- Soil-Water Modelling and Decision Support through a Graphical User Interface for the Daisy Model